

Non-WEP coefficients destroy free trigonometric factorization

A counterexample to the C*-coefficient extension asked in arXiv:2511.06487

Automated mathematical-discovery candidate (lane 11)

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Status. Candidate counterexample; current verdict **likely valid**. It completely answers the paper's broad question negatively by refuting the trigonometric theorem for every non-WEP coefficient algebra. It does not decide the separate unconstrained self-adjoint-polynomial extension.

1 Source problem

Jindal, Klep, and McCullough prove in Theorem 1.3 that a $\mathcal{B}(\mathcal{H})$ -valued trigonometric polynomial on the free group which is positive under every spatial unitary evaluation is a finite sum of hermitian squares (and, in their infinite-dimensional setting, a single square) [1]. On PDF page 22, Section 8, they ask whether their results remain true for arbitrary C*-algebra coefficients:

GNS argument yields Theorem 1.3. □

8. CONCLUDING REMARK AND A PROBLEM

The reader will have no difficulty verifying that both Theorem 1.1 and Theorem 1.3 remain true if we replace the algebra of coefficients $\mathcal{B}(\mathcal{H})$ with a von Neumann algebra. The proofs carry over to this setting in a straightforward way. On the other hand, we do not know if the results still hold if the coefficients are from a C*-algebra.

DECLARATIONS

Competing interests. The authors have no relevant financial or non-financial interests to disclose.

Data availability. This article is theoretical in nature; no datasets were generated or analyzed. All results and proofs are contained within the article.

We keep exactly the source's spatial evaluation convention. Thus, for a faithful representation $\pi : \mathcal{A} \rightarrow \mathcal{B}(H)$ and $p = \sum_{w \in \mathbb{F}_2} a_w w \in \mathcal{A}[\mathbb{F}_2]$, evaluation at a unitary pair $U = (U_1, U_2)$ on K is

$$p(U) = \sum_w \pi(a_w) \otimes U^w \in \mathcal{B}(H \otimes K).$$

The result below shows that the answer to the broad question is no.

2 Proof intuition

Spatial evaluations see the *minimal* tensor product $\mathcal{A} \otimes_{\min} C^*(\mathbb{F}_2)$. In contrast, an algebraic sum of hermitian squares is positive in *every* C*-completion, hence in the maximal tensor product. For an

algebra without the weak expectation property (WEP), those two tensor norms differ. Choose a finite group-ring tensor x whose maximal norm is larger than its minimal norm, and place a scalar λ strictly between their squares. Then

$$p = \lambda 1 - x^* x$$

is strictly positive in the minimal completion, so every spatial unitary evaluation is positive, but it is not positive in the maximal completion. This rules out not only a single square but every finite sum of squares.

3 The WEP input

We use the following theorem of Farenick, Kavruk, and Paulsen [2, Theorem 4.3]. For every unital C^* -algebra $\mathcal{A}$,

$$\mathcal{A} \otimes_{\min} C^*(\mathbb{F}_2) = \mathcal{A} \otimes_{\max} C^*(\mathbb{F}_2) \iff \mathcal{A} \text{ has WEP.} \quad (1)$$

Their statement (take $n = 3$) is reproduced below.

Proposition 8.8] there is a ucp projection $\psi : C^*(\mathbb{F}_n) \rightarrow C^*(\mathbb{F}_n)$ with range $C^*(\mathbb{F}_\infty)$ (considered as a unital C^* -subalgebra of $C^*(\mathbb{F}_n)$). Thus,

$$\mathcal{A} \otimes_{\min} C^*(\mathbb{F}_\infty) \subset \mathcal{A} \otimes_{\min} C^*(\mathbb{F}_n) = \mathcal{A} \otimes_{\max} C^*(\mathbb{F}_n) \rightarrow \mathcal{A} \otimes_{\max} C^*(\mathbb{F}_\infty)$$

yields a ucp map $\mathcal{A} \otimes_{\min} C^*(\mathbb{F}_\infty) \rightarrow \mathcal{A} \otimes_{\max} C^*(\mathbb{F}_\infty)$, which implies that $\mathcal{A} \otimes_{\min} C^*(\mathbb{F}_\infty) = \mathcal{A} \otimes_{\max} C^*(\mathbb{F}_\infty)$. This proves the implication $(1) \Rightarrow (3)$. But using the fact that any two countable free groups are subgroups of each other, the same arguments apply to obtain the equivalence of statements (1) , (2) , and (3) . $\square$

Theorem 4.3. *For a fixed $n \geq 2$, the following statements are equivalent for a unital C^* -algebra $\mathcal{A}$.*

- (1) $\mathcal{A}$ has $(\mathfrak{S}_n)$.
- (2) $\mathcal{A} \otimes_{\min} \mathcal{S}_{n-1} = \mathcal{A} \otimes_{\max} \mathcal{S}_{n-1}$.
- (3) $\mathcal{A} \otimes_{\min} C^*(\mathbb{F}_{n-1}) = \mathcal{A} \otimes_{\max} C^*(\mathbb{F}_{n-1})$.

If $n = 2$, then the equivalent statements (1) , (2) , and (3) hold for every $\mathcal{A}$. If $n \geq 3$, then the equivalent statements (1) , (2) , and (3) hold if and only if $\mathcal{A}$ has WEP.

Proof. $(1) \Leftrightarrow (2)$ The complete quotient map $\phi : \mathcal{T}_n \rightarrow \mathcal{S}_{n-1}$ induces a complete quotient map $\text{id}_{\mathcal{R}} \otimes \phi : \mathcal{R} \otimes_{\max} \mathcal{T}_n \rightarrow \mathcal{R} \otimes_{\max} \mathcal{S}_{n-1}$ for every operator system $\mathcal{R}$ [6 Proposition 1.6]. Such is the case in particular for $\mathcal{R} = \mathcal{A}$. Because $\otimes_{\min} = \otimes_c$ if one of the tensor factors is $\mathcal{T}_n$ [6 Proposition 4.1] and because $\otimes_c = \otimes_{\max}$ if one of the tensor factors is a unital C^* -algebra [8 Theorem 6.6], we deduce that $\mathcal{A} \otimes_{\min} \mathcal{T}_n = \mathcal{A} \otimes_{\max} \mathcal{T}_n$. Now consider the following commutative diagram:

$$\begin{array}{ccc} \mathcal{A} \otimes_{\min} \mathcal{T}_n & \xrightarrow{\cong} & \mathcal{A} \otimes_{\max} \mathcal{T}_n \\ \text{id}_{\mathcal{A}} \otimes \phi \downarrow & & \downarrow \text{id}_{\mathcal{A}} \otimes \phi \\ \mathcal{A} \otimes_{\min} \mathcal{S}_{n-1} & \xrightarrow{\theta = \text{id}} & \mathcal{A} \otimes_{\max} \mathcal{S}_{n-1} \end{array}$$

The top arrow is a complete order isomorphism, the rightmost arrow is a complete quotient map, while the bottom arrow is a linear isomorphism θ in which θ^{-1} is completely positive. By [6 Lemma 5.1], $\text{id}_{\mathcal{A}} \otimes \phi : \mathcal{A} \otimes_{\min} \mathcal{T}_n \rightarrow \mathcal{A} \otimes_{\min} \mathcal{S}_{n-1}$ is a complete quotient map if and only if θ is a complete order isomorphism. But since $\text{id}_{\mathcal{A}} \otimes \phi : \mathcal{A} \otimes_{\min} \mathcal{T}_n \rightarrow \mathcal{A} \otimes_{\min} \mathcal{S}_{n-1}$ is a complete quotient map if and only if $\mathcal{A}$ has $(\mathfrak{S}_n)$ (by Theorem 2.1(2)), we deduce the equivalence $(1) \Leftrightarrow (2)$.

$(2) \Rightarrow (3)$. By Lemma 4.1, $\mathcal{A} \otimes_c \mathcal{S}_{n-1} \subset_{\text{coi}} \mathcal{A} \otimes_c C^*(\mathbb{F}_{n-1})$. Thus, $\mathcal{A} \otimes_{\max} \mathcal{S}_{n-1} \subset \mathcal{A} \otimes_{\max} C^*(\mathbb{F}_{n-1})$ since $\otimes_c = \otimes_{\max}$ if one of the tensor factors is a C^* -algebra. By the

The equality in (1) means equality of the two C^* -norms on the algebraic tensor product. Therefore, if $\mathcal{A}$ lacks WEP, some algebraic tensor has distinct norms.

4 Counterexample theorem

Theorem 1. *Let $\mathcal{A}$ be a unital C^* -algebra without WEP. There exist an integer d , a finite element $x \in \mathcal{A} \odot \mathbb{C}[\mathbb{F}_2]$ supported on words of length at most d , and a self-adjoint trigonometric polynomial*

$$p = \lambda 1 - x^* x \in \mathcal{A}[\mathbb{F}_2]$$

of degree at most $2d$ such that:

1. $p(U) > 0$ for every unitary pair U on every Hilbert space;
2. p is not a finite sum of hermitian squares in $\mathcal{A}[\mathbb{F}_2]$.

In particular, the C^ -coefficient extension of Theorem 1.3 of [1] is false.*

Proof. By (1), the minimal and maximal norms on $\mathcal{A} \odot C^*(\mathbb{F}_2)$ differ. Hence there is $y = \sum_{j=1}^m a_j \otimes b_j$ with $\|y\|_{\max} > \|y\|_{\min}$. Approximate each b_j in norm by $c_j \in \mathbb{C}[\mathbb{F}_2]$. For either tensor norm $\alpha \in \{\min, \max\}$,

$$\left\| \sum_j a_j \otimes (b_j - c_j) \right\|_{\alpha} \leq \sum_j \|a_j\| \|b_j - c_j\|.$$

Choosing the approximations sufficiently close preserves the strict gap. Thus, after replacing y , we have a finite $x \in \mathcal{A} \odot \mathbb{C}[\mathbb{F}_2]$ with

$$\|x\|_{\min} < \|x\|_{\max}. \quad (2)$$

Choose a real number

$$\|x\|_{\min}^2 < \lambda < \|x\|_{\max}^2 \quad (3)$$

and set $p = \lambda 1 - x^* x$.

In the minimal tensor product,

$$p \geq (\lambda - \|x\|_{\min}^2) 1 > 0.$$

Every unitary pair U induces a unital $*$ -representation $\rho_U : C^*(\mathbb{F}_2) \rightarrow \mathcal{B}(K)$. The spatial evaluation of the source is the image of p under $\pi \otimes \rho_U$ from the minimal tensor product. It follows that

$$p(U) \geq (\lambda - \|x\|_{\min}^2) I_{H \otimes K} > 0.$$

Suppose, toward a contradiction, that $p = \sum_{k=1}^N r_k^* r_k$ with $r_k \in \mathcal{A}[\mathbb{F}_2]$. Under the canonical map into $\mathcal{A} \otimes_{\max} C^*(\mathbb{F}_2)$, the right-hand side is positive. Hence $\lambda 1 - x^* x \geq 0$ in that C^* -algebra. Positivity would imply $x^* x \leq \lambda 1$, and therefore $\|x\|_{\max}^2 \leq \lambda$, contradicting (3). Thus no finite sum-of-squares factorization exists. $\square$

Corollary 1. *If the two-variable trigonometric factorization conclusion of Theorem 1.3 in [1] holds for a unital coefficient algebra $\mathcal{A}$ under spatial evaluations, then $\mathcal{A}$ has WEP.*

Proof. This is the contrapositive of Theorem 1. $\square$

Non-WEP algebras exist. For example, a von Neumann algebra has WEP exactly when it is injective [2, Corollary 6.2]; thus a standard noninjective von Neumann algebra such as the free group factor $L(\mathbb{F}_2)$ supplies a coefficient algebra in Theorem 1. Under the same spatial evaluation convention, this also shows that the source's preceding blanket von Neumann-algebra remark needs an injectivity (or a different positivity- semantics) qualification.

5 Scope, verification, and novelty

The construction answers the source’s broad question without an explicit closed formula for x : failure of WEP supplies an algebraic norm-gap witness, and density moves that witness into the finite group ring. The proof has no computational dependency. Verification checked the direction of both tensor norm inequalities, the preservation of the gap under group-ring approximation, strict positivity of every spatial evaluation, and the fact that any finite sum of squares is maximal-positive.

The separate extension of the source’s unconstrained self-adjoint-polynomial Theorem 1.1 is not decided here. Cayley-transform and symmetry-dilation attempts produce rational denominators or positivity only on constrained tuples; no exact implication back from polynomial squares to group-ring squares was obtained. Nor is WEP claimed sufficient for exact algebraic factorization: $\min=\max$ removes this obstruction, but positivity in a completed tensor product alone gives no bounded-degree algebraic square certificate.

A bounded search on 12 August 2026 covered the exact title and arXiv id, **C*-algebra coefficients**, **WEP**, **operator-valued noncommutative trigonometric polynomial**, and their combinations. It found the source and general WEP/tensor-product literature, but no later answer to this concluding problem or this norm-gap counterexample. Novelty is plausible, not certified. Human review should focus on the intended spatial evaluation convention and the application of [2, Theorem 4.3] with $n = 3$.

References

- [1] A. Jindal, I. Klep, and S. McCullough, *Positive operator-valued noncommutative polynomials are squares*, arXiv:2511.06487; Integral Equations Operator Theory 98 (2026), Article 7, especially Theorem 1.3 and PDF page 22.
- [2] D. Farenick, A. S. Kavruk, and V. I. Paulsen, *C*-algebras with the weak expectation property and a multivariable analogue of Ando’s theorem on the numerical radius*, arXiv:1107.0418, especially Theorem 4.3 and Corollary 6.2.